



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SAKTHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

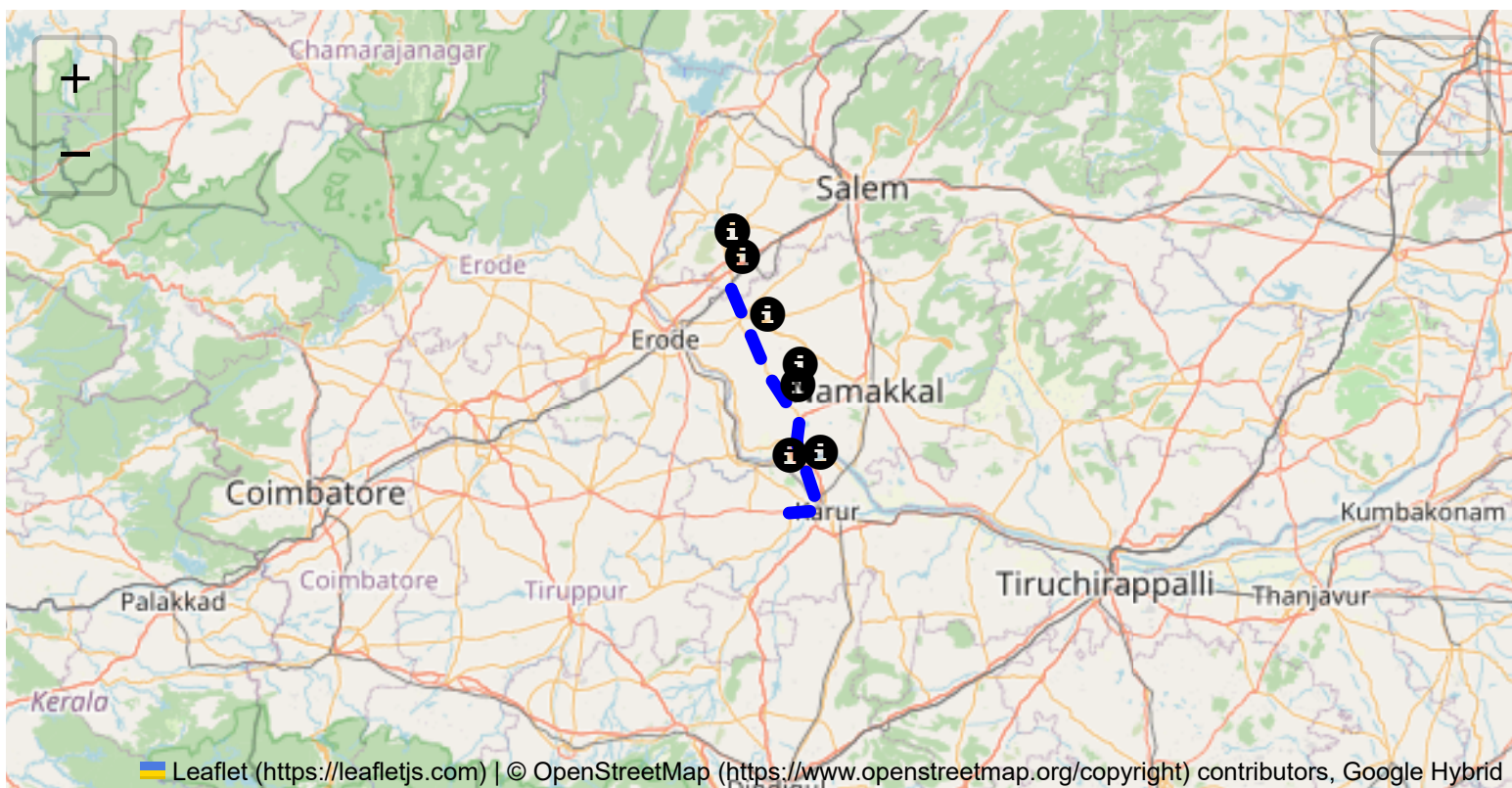
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

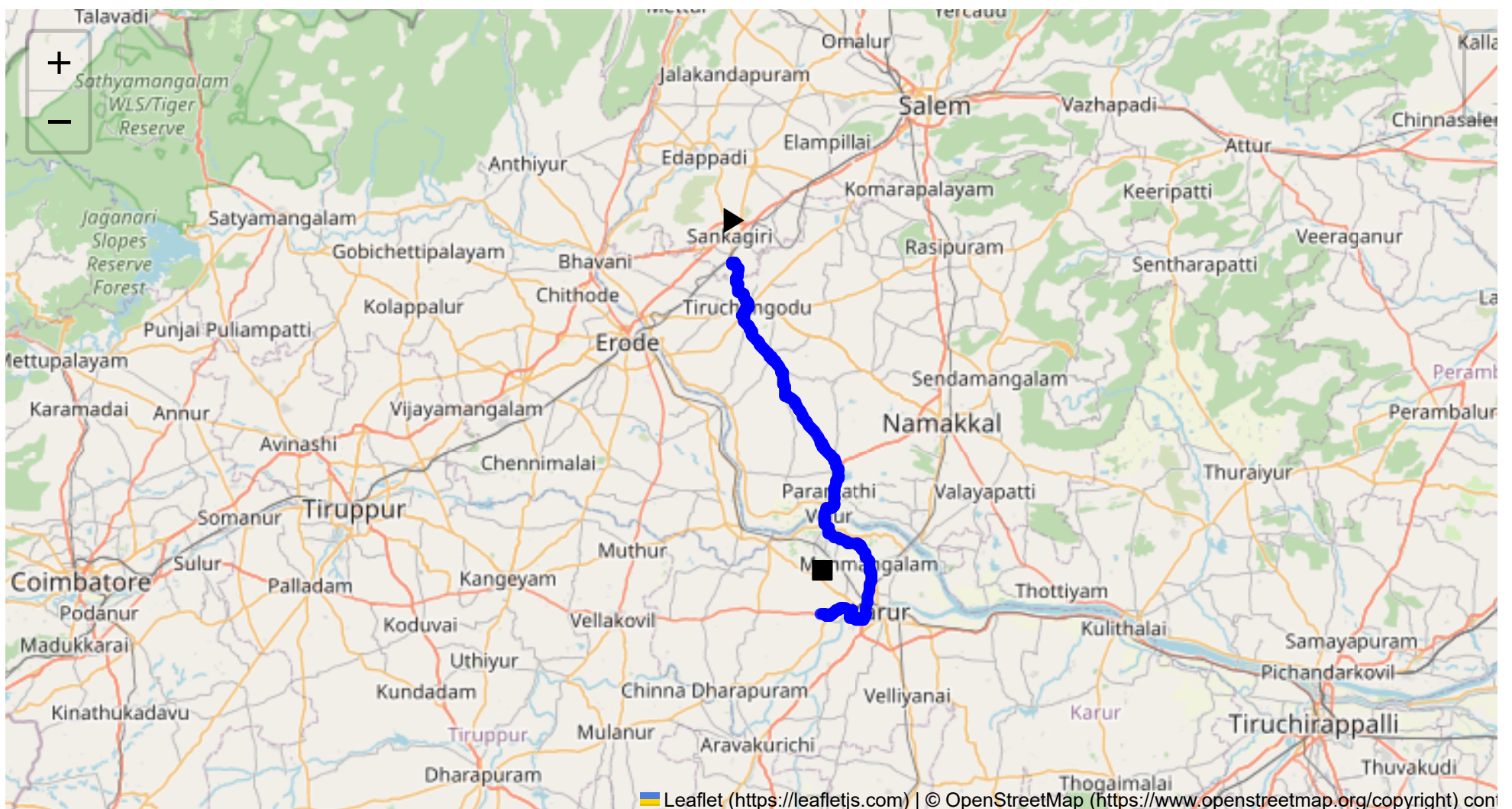
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 74.25 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (10.956731, 77.996517)



Welcome to the Journey Risk Management Study

Route Overview

The route stretches approximately 74.25 kilometers, starting in Sangagiri and terminating in Pavitiramedu, Tamil Nadu. This journey encompasses various state highways and bypass roads, passing through towns like Puthur, Tiruchengode, Musalnaickenpalayam, Paramathi, Velur, and Karur.

Weather Conditions and Hazards

The climate in Tamil Nadu can vary, but it generally experiences hot and humid conditions. Monsoons are a prominent weather hazard, with rains common between October and December. These can lead to slippery roads and waterlogging in low-lying areas. During peak summer months, high temperatures may also cause road surfaces to become sticky or even degrade.

Traffic Patterns

- **Peak Hours:** Typically, traffic congestion is observed during school timings (7:30-9:00 AM and 3:00-4:30 PM) and standard commuting hours (8:00-10:00 AM and 5:00-7:00 PM).
- **Congestion-Prone Areas:** Particularly heavy congestion can occur near schools and major intersections, such as near the Girls High School in Tiruchengode and along the Karur-Namakkal Highway.

Road Quality and Infrastructure

Overall, the route is serviced by state highways and bypass roads, which are generally well-maintained. However, some segments, particularly in rural areas, may have narrow lanes, potholes, or wear due to heavy vehicle use. Road signage may not be adequately illuminated in less populated zones.

Alternative Routes for Emergencies

- Using alternate state highways that run parallel can be beneficial during road blockages.
- Local district roads may provide detours, although they may be less maintained.

Regulations on Hazardous Material Transport

Local transport regulations require that vehicles carrying hazardous materials comply with the state's guidelines, which include designated routes and specific timings to minimize risk in populated areas. Timely intimation to local authorities is also recommended.

Historical Incidents

There are occasional reports of accidents involving heavy vehicles, often attributed to over-speeding or evasive maneuvers due to unexpected road conditions. Incidents involving hazardous materials without significant environmental impact are rare but require vigilance.

Environmental Considerations

Ensure careful travel through residential and agricultural zones to avoid contamination from hazardous materials. Minimal impact on wildlife or reserved natural areas is expected but maintaining vigilance is crucial.

Communication Coverage

Mobile network coverage is generally reliable, but potential dead zones may exist in more rural stretches, especially between Musalnaickenpalayam and Paramathi. Access to reliable GPS signals may fluctuate in these areas.

Emergency Response Times

Depending on proximity to urban centers or major towns, emergency response times may vary from approximately 20 minutes in urban areas to over an hour in more remote locations.

Overall Summary of Risk Assessment

This route is comparatively safe for hazardous material transport, with manageable risks pertaining to traffic congestion, weather conditions, and road quality. Adequate planning around weather events and adherence to local regulations should ensure safe transit. Informing local authorities of travel plans and preparing for variable communication coverage will further mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44022, 77.87546	15 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16755, 78.01568	30 KM/Hr
7	Turn	Medium	11.15579, 78.02169	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	10.95094, 78.05884	15 KM/Hr
11	Turn	Medium	10.95026, 78.04855	30 KM/Hr
12	Turn	High	10.95212, 78.03978	15 KM/Hr
13	Blind Spot	Blind Spot	10.96305, 78.04058	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium
20	school	st. antonys matric hr. secondary school	10.9641269, 78.0417919	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15579, 78.02169



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95026, 78.04855



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.95212, 78.03978



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 10.96305, 78.04058

